A Comparative Analysis of Capacitor-Less LDO Topologies in 130nm CMOS Technology

Kourosh Nourmohammadi Baygi

Circuits and Systems Group
School of Electrical and Computer Engineering
National Technical University of Athens

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Outline

1. Introduction
2. Proposed Architectures
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Motivation & Target Specifications

Design Goal: Implement a stable, capacitor-less PMOS Low-Dropout Regulator in 130 nm CMOS technology.

Parameter	**Target**
Supply Voltage (V_{EXT})	3 V
Output Voltage (V_{OUT})	2.3 V
Max Load Current (I_{load})	500 µA
Phase Margin (Full/No Load)	$> 70°/ > 60°$

Error Amplifier Topologies

To achieve the desired gain and PSRR without an external capacitor, three OTA topologies were evaluated:

- **Standard 5T OTA:** Baseline for area and current consumption.
- **Telescopic OTA:** High gain and bandwidth, but limited output swing.
- **Current Mirror OTA:** Improved slew rate and decoupling of bandwidth from the input pair.

Sizing Methodology

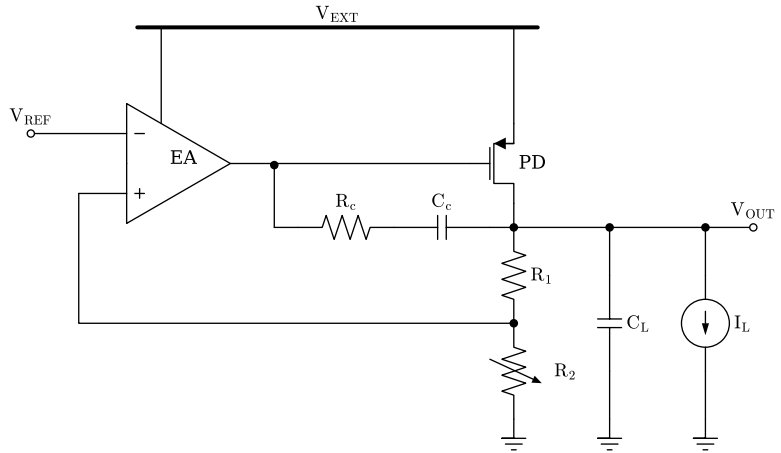


Figure:

Telescopic OTA

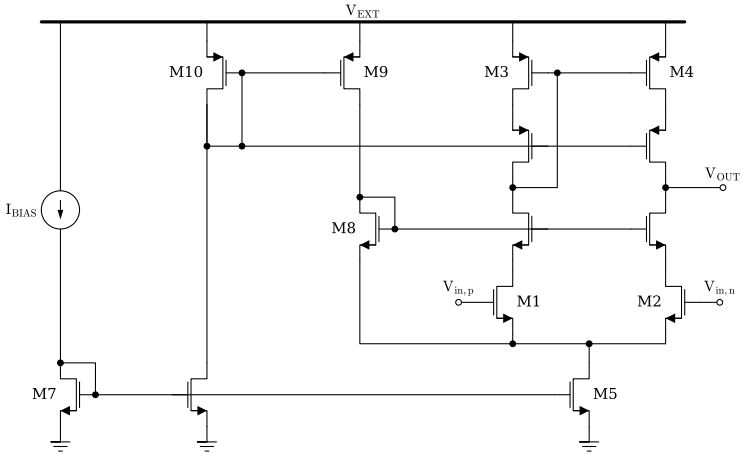


Figure:

Bode Plot

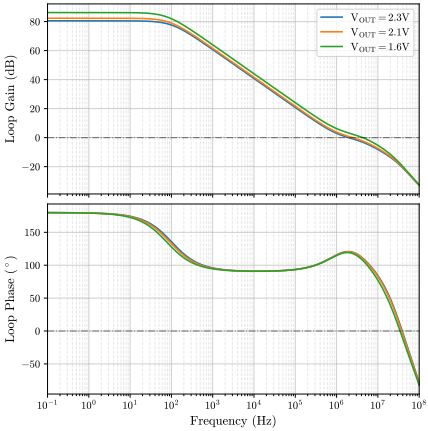


Figure:

Transient Response & PSRR

Load Transient Response

- Settling time (t_s): $< 40\,\text{ns}$
- Undershoot/Overshoot within acceptable margins.

Power Supply Rejection Ratio

- Low-frequency PSRR: $> 65\,\text{dB}$
- High-frequency PSRR (1 MHz): $> 2\,\text{dB}$

Conclusion

- The **Telescopic OTA** provided the best overall noise performance.
- The **Current Mirror OTA** offered superior dynamic response at the cost of slight area increase.
- All topologies successfully met the capacitor-less stability criteria across the specified load range.

Thank you!

Questions?